Anand Bhattad

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EDUCATION

University of Illinois Urbana-Champaign (UIUC), USA

Ph.D., Computer Science — *Jan'19 - May'24*
Thesis: Exploring Knowledge in Generative Models
PhD Defense: May'23; *Degree awarded*: May'24
Thesis committee: David Forsyth (Advisor), Alyosha Efros, Bill Freeman, Derek Hoiem, Svetlana Lazebnik and Shenlong Wang

M.S., Computer Science — *Aug'17 - Dec'18*
M.S., Civil Engineering — *Aug'15 - Aug'17*

National Institute of Technology Karnataka (NITK) Surathkal, India

B.Tech., Civil Engineering — *July'11 - May'15*

RESEARCH EXPERIENCE

Toyota Technological Institute at Chicago, Chicago — *Sept'23 - present*
Research Assistant Professor

University of California Berkeley, Berkeley — *Starting May'24*
Visiting Scholar; Host: *Alyosha Efros*

Allen Institute of Artificial Intelligence, Seattle/Champaign(remote) — *Feb'23 - April'23*
Research Intern with *Ani Kembhavi*

Intel ISL, Santa Clara/Champaign (remote) — *Feb'21 - Aug'21*
Research Intern with *Stephan R. Richter* in *Vladlen Koltun's* team.

NVIDIA ADLR, Santa Clara/Champaign (remote) — *May'20 - Dec'20*
Research Intern with *Andrew Tao and Bryan Catanzaro*

Fyusion Inc, San Francisco — *May'19 - Aug'19*
Research Intern with *Abhishek Kar and Rodrigo Ortiz Cayon*

Siemens Corporate Technology, Princeton — *May'17 - Aug'17*
Computer Vision & DL Intern with *Jan Ernst*

Computational Sensor Lab, Michigan State University — *May'14 - July'14*
Research Intern with *Prof. Nizar Lajnef*

Earthquake Engineering Research Center, IIIT Hyderabad — *Jun'13 - July'13*
Research Intern with *Prof. Ramacharla Pradeep Kumar*

CONFERENCE PAPERS

10. ***Shadows Don't Lie and Lines Can't Bend!* Generative Models don't know Projective Geometry...for now**
A. Sarkar*, H. Mai*, A. Mahapatra*, S. Lazebnik, D.A. Forsyth, **A. Bhattad**
CVPR 2024
9. **StyLitGAN: Image-based Relighting via Latent Control**
A. Bhattad, J. Soole, D. A. Forsyth
CVPR 2024
8. **StyleGAN Knows Normals, Depth, Albedo, and More**
A. Bhattad, D. McKee, D. Hoiem, D. A. Forsyth
NeurIPS 2023

7. **OBJect 3DIT: Language-guided 3D-aware Image Editing**
   O. Michel, **A. Bhattad**, E. Vanderbilt, R. Krishna, A. Kembhavi, T. Gupta
   NeurIPS 2023
6. **Improving Equivariance in State-of-the-Art Supervised Depth and Normal Predictors**
   Yuanyi Zhong, **A. Bhattad**, Y. Wang, D. A. Forsyth
   ICCV 2023
5. **Cut-and-Paste Object Insertion by Enabling Deep Image Prior for Reshading**
   A. Bhattad, D. A. Forsyth
   3DV 2022 (Spotlight Presentation)
4. **DIVeR: Real-time and Accurate Neural Radiance Fields with Deterministic Integration for Volume Rendering**
   L. Wu , J. Y. Lee, **A. Bhattad**, Y. Wang, D. A. Forsyth
   CVPR 2022 (Oral Presentation) **(Best Paper Finalist)**
3. **View Generalization for Single Image Textured 3D Models**
   A. Bhattad, A. Dundar, G. Liu, A. Tao, B. Catanzaro
   CVPR 2021
2. **Unrestricted Adversarial Perturbations via Semantic Manipulation**
   A. Bhattad*, M. J. Chong*, K. Liang, B. Li, D. A. Forsyth (* for equal contribution)
   ICLR 2020
1. **Improved Style Transfer with Calibrated Metrics**
   M. Yeh*, S. Tang*, **A. Bhattad**, C. Zou, D. A. Forsyth (* for equal contribution)
   WACV 2020

Workshop Papers

3. **MIMIC: Masked Image Modeling with Image Correspondences**
   K. Marathe, M. Bigverdi, N. A. Khan, T. Kundu, P. Howe, S. Ranjit, **A. Bhattad**, A. Kembhavi, L. Shapiro, R. Krishna
   CVPR-W 2024
2. **Big but Imperceptible Adversarial Perturbations via Semantic Manipulation**
   A. Bhattad*, M. J. Chong*, K. Liang, B. Li, D. A. Forsyth (* for equal contribution)
   CVPR-W 2019, ***Oral Presentation***
1. **Detecting Anomalous Faces with "No Peeking" Autoencoders**
   A. Bhattad, J. Rock, D. A. Forsyth
   CVPR-W, 2018, ***Oral Presentation***

Under Review

4. **Intrinsic LoRA: A Generalist Approach for Discovering Knowledge in Generative Models**
   X. Du, N. Kolkin, G. Shakhnarovich, **A. Bhattad**
   In Submission
3. **Videoshop: Localized Semantic Video Editing with Noise-Extrapolated Diffusion Inversion**
   X. Fan, **A. Bhattad***, R. Krishna*
   In Submission
2. **Intrinsic ControlNet: Zero-shot Object Compositing from Image Intrinsics**
   Z. Zhang, F. Fortier-Chouinard, M. Garon, **A. Bhattad**, JF. Lalonde
   In Submission
1. **UrbanIR: Large-Scale Urban Scene Inverse Rendering from a Single Video**
   Z. Lin, B. Liu, Y. Chen, D.A. Forsyth, J. Huang, **A. Bhattad**, S. Wang
   In Submission

Technical Reports

3. **Blocks2World: Controlling Realistic Scenes with Editable Primitives**
   V. Vavilala, S. Jain, R. Vasanth, **A. Bhattad**, D.A. Forsyth
   Technical Report, arXiv 2023
2. **Make It So: Steering StyleGAN for Any Image Inversion and Editing**
   A. Bhattad, V. Shah, D. Hoiem, D. A. Forsyth
   Technical Report, arXiv 2023

1. **SIRfyN: Single Image Relighting from your Neighbors**
   D. A. Forsyth, **A. Bhattad**, P. Asthana, Y. Zhong, Y. Wang
   Technical Report, arXiv 2021

Students Mentoring

Doctoral Students

- Xiao Zhang (University of Chicago; 2023-present) | Project on Latent Intrinsics
- Xiaodan Du (TTI-Chicago; 2023-present) | Paper Under Review (see UR#4)
- Xiang Fan (University of Washington; 2023-present) | Paper Under Review (see UR#3)
- Zitian Zhang (Université Laval; 2023-present) | Paper Under Review (see UR#2)
- Ayush Sarkar (UIUC; 2023-present) | Paper at CVPR (see CP#10)
- Kalyani Marathe (University of Washington; 2023-present) | Paper at CVPR-W (see WP#1)
- Zhi-Hao Lin (UIUC; 2023-present) | Paper Under Review (see UR#1)

Masters Students

- Hanlin (Asher) Mai (UIUC; 2023-present) | Paper at CVPR (see CP#10)
- Seemandhar Jain (UIUC; 2023-2024) | Project on Latent Intrinsics | Next: PhD Student at UCSD
- Frédéric Fortier-Chouinard (Université Laval; 2023-present) | Paper Under Review (see UR#2)
- James Soole (UIUC; 2023) | Paper at CVPR (see CP#9)
- Feiran Wang (UIUC; 2023) | Project on NeRF + Illumination Physics
- Kexuan (Klaus) Zou (UIUC; 2019-2020) | Next: Software Engineer at NVIDIA

Bachelors Students

- Joshua Ahn (University of Chicago; 2024-present) | Project on NeRF w/o Positional Encoding
- Amitabh Mahapatra (UIUC; 2023-present) | Paper at CVPR (see CP#10)
- Kuan-Sheng Chen (UIUC; 2023) | Project on Generative Models with Intrinsic Images
- Oscar Michel (AI2; 2023) | Paper at NeurIPS (see CP#7) | Next: PhD Student at NYU
- Liwen Wu (UIUC; 2022) | Paper at CVPR (see CP#4) | Next: PhD Student at UCSD
- Brian Chen (UIUC; 2020-2022) | REU at CMU's Robotics Institute
- Anchu Zhu (UIUC; 2018-2019) | Next: MS in CS at USC

Teaching Experience

Past Meets Present: A Tale of Two Visions, TTIC; *Course Instructor* Spring 2024

- Designed and developed a new course on Computer Vision, combining historical and modern approaches
- Taught a class of 20+ students from TTIC and UChicago, fostering an engaging learning environment

Computer Science, UIUC; *Graduate Teaching Assistant* Jan 2016 – Dec 2018

- Applied Machine Learning (CS498 AML), Fall 2018
  - Designed and implemented Kaggle competitions to enhance student learning and engagement
- Data Structures (CS 225), Spring 2017
  - Developed assignments, exams, and labs to assess and reinforce student understanding
- Introduction to Computing (CS 101), Spring 2016 & Fall 2016
  - Recognized as an **Excellent TA** for effective teaching and student support

Invited Talks

What do Generative Image Models Know?

- IIIT Hyderabad, India; Vision Seminar Jan 2024
- TTI-Chicago; Research@TTIC Oct 2023
- Exactech, Inc.; Tech Talk Oct 2023

Exploring Knowledge in Generative Models

- Stanford University in Jiajun Wu's group June 2023
- University of Tubingen, Autonomous Vision Group May 2023

What do Generative Models know about Geometry and Illumination?

- UC Berkeley: Vision Seminar Apr 2023
- NVIDIA Research Apr 2023
- MIT: Vision and Graphics Seminar Apr 2023
- CMU: VASC Seminar Mar 2023

- UW: Vision Seminar — Mar 2023

Learning about Light without Labeled Data

- UMD: Vision Seminar — Mar 2023
- UCSD: Pixel Cafe Seminar — Feb 2023
- TTIC: Research Talk — Feb 2023

Services & Leadership

Lead Organizer, ECCV-W on "Knowledge in Generative Models" — Sept 2024
Lead Organizer, CVPR-W on "CV 20/20: A Retrospective Vision" — June 2024
Lead Organizer, CVPR-W on "Scholars & Big Models: How Can Academics Adapt?" — June 2023
Lead Organizer, UIUC Vision Workshop at Allerton — April 2023
Organizer, UIUC External Speaker Series and Vision Lunch — Jan 2021 – May 2023
Reviewer, Computer Vision Conferences — Nov 2018 – Present

- Serving as reviewer for CVPR, NeurIPS, ECCV and ICCV conferences and workshops.
- **Outstanding Reviewer** for ICCV 2023
- **Outstanding Emergency Reviewer** for CVPR 2021

Program Committee — Jun 2019

- Adversarial Machine Learning in Real-World Computer Vision Systems, CVPR 2019 Workshop
- Security and Privacy in Machine Learning, ICML 2019 Workshop

Organizer, Vision Coffee, Urbana-Champaign — *May'18 - Aug'18*

- Vision and Deep Learning paper/topic weekly discussion group
- Finalized agenda/topic, location and initiated discussions

Co-Founder, SchoolEngg — *Apr'13 - May'15*

- Forum for high-school students; providing firsthand knowledge of all engineering disciplines
- Currently, renamed as *PrepLift*, an education counselling start-up in India